

Digital Electronics Lab Report

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Date of experiment : 24-3-26

Experiment 9: Simulating UART communication

Objectives:

1. To send 4-bit data from breadboard to PC using UART protocol.
2. To understand serial communication using start bit, parity bit, and stop bit.

Apparatus:

- IC-7486 Quad XOR Gate (1)
- IC-74165 PISO Register (1)

- Resistor 1 k Ω (4)
- Resistor 100 k Ω (2)
- Capacitor 100 nF (1)
- Push Button (1)
- Toggle switch / DIP switch (1)
- Breadboard
- DIP Switch (1)
- Connecting wires
- DC Power Supply 5V (1)
- PC or Laptop (1)
- USB to Serial Converter (1)

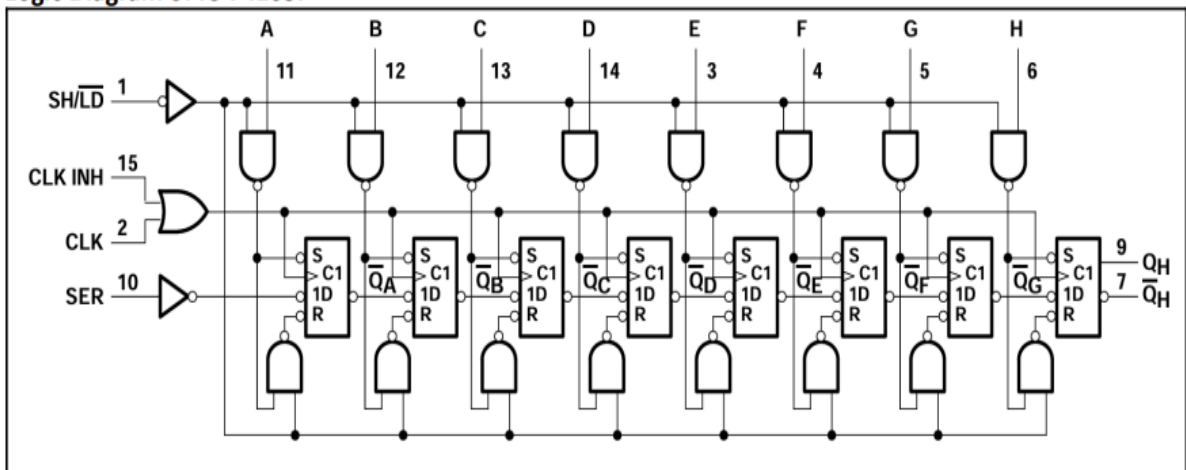
Theory:

UART (Universal Asynchronous Receiver Transmitter) is a serial communication protocol used for transmitting data between devices. In UART communication, data is sent bit by bit over a single line along with control bits such as start bit, parity bit, and stop bit. In this experiment, a 4-bit parallel input is converted into serial form using a PISO (Parallel In Serial Out) register IC-74165. The XOR gate

(IC-7486) is used to generate parity for error detection. The serial output is transmitted to a PC using a USB to serial converter. The communication is carried out at a baud rate of 9600 with 1 start bit, 1 parity bit, and 1 stop bit. This experiment demonstrates the basic working of serial data transmission and interfacing between digital hardware and a computer.

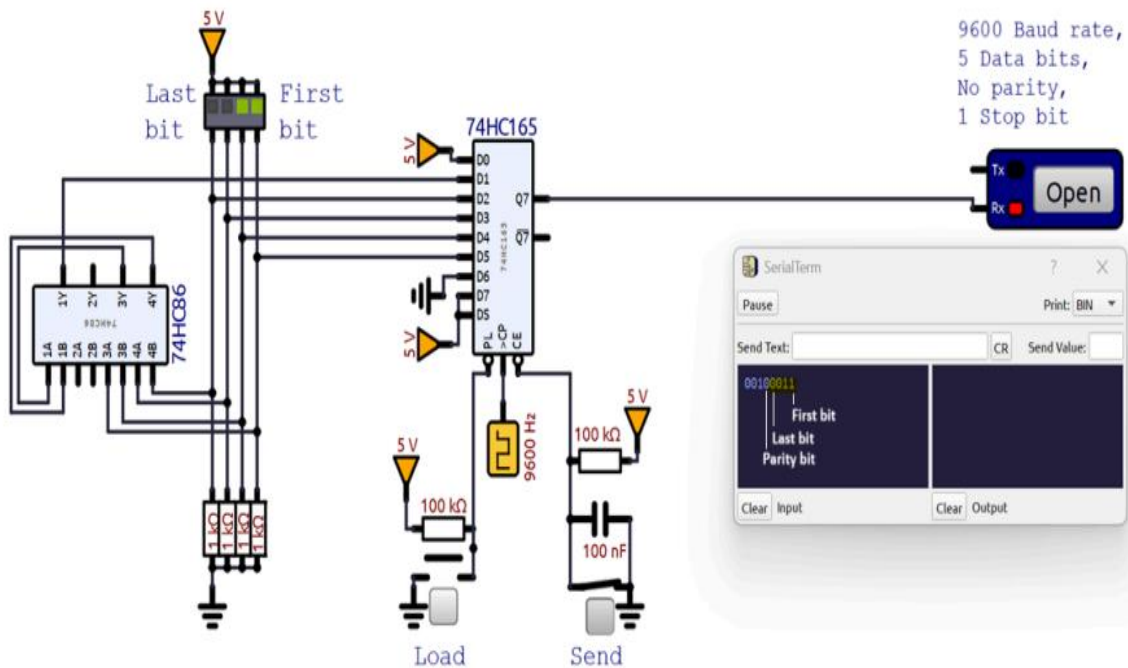
IC-74165 (Parallel-In Serial-Out Register)

Logic Diagram of IC-74165:



The IC-74165 is an 8-bit Parallel-In Serial-Out (PISO) shift register used to convert parallel data into serial form. It is widely used in digital communication systems where multiple bits must be transmitted over a single data line.

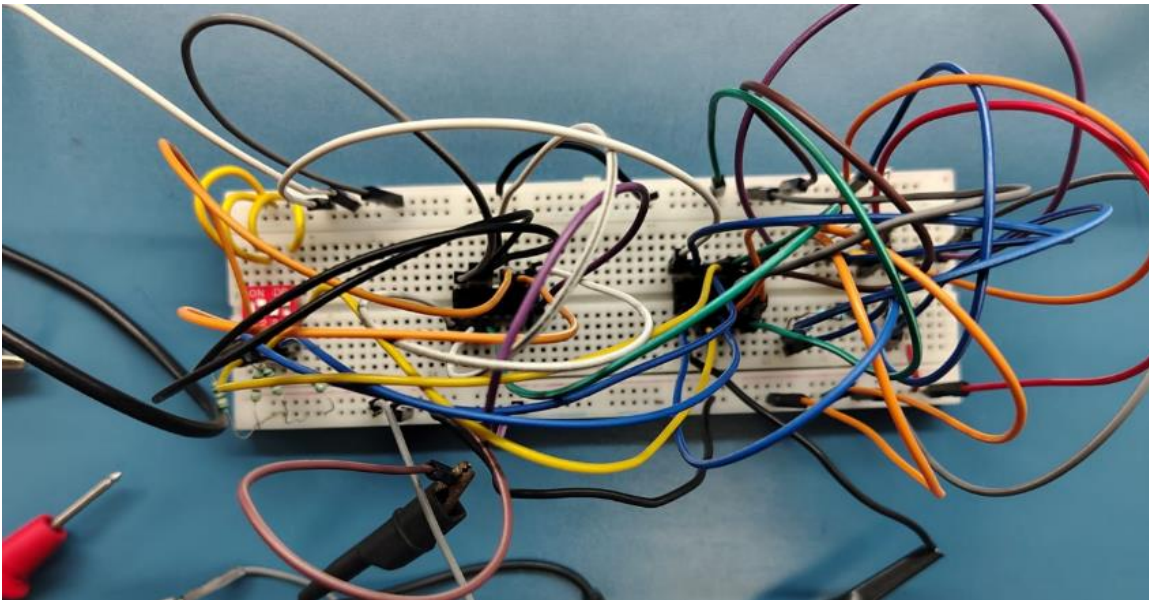
Circuit Diagram:



Output/Observation:

1. The 4-bit input was successfully loaded into the PISO register.
2. Upon triggering, the data was transmitted serially to the PC.

3. The received data matched the transmitted input values.
4. Proper framing with start, parity, and stop bits was observed.
5. Communication was stable at 9600 baud rate.



Conclusion:

The UART communication between the breadboard circuit and PC was successfully implemented. The experiment demonstrated the conversion of parallel data into serial form and its reliable transmission using UART protocol. It provided practical understanding of serial communication, data framing, and interfacing techniques used in digital systems.